



Global AI Trend Tracker

Global Markets Research

22 September 2026

EQUITY: TECHNOLOGY

AI expert call 73: optical comm sector update – electrical chips

Quick Note

We hosted an AI Expert Call #73 on 22 September, featuring an expert from an AI infra service provider. The expert mainly discussed the supply and demand dynamics of 800G/1.6T DSP (Digital Signal Processor), technology barriers and supply bottlenecks, competitive landscape of the DSP industry, pricing trend for DSP, TIA (Transimpedance Amplifier) and drivers, as well as product progress of TIA and drivers. Key takeaways from the call are below:

800G / 1.6T DSP supply and demand dynamics

- In 2025, about 25-26mn units of 800G DSP and over 3mn units of 1.6T DSPs were shipped, according to the expert.
- For 2026E, the expert estimates that close to 50mn units of 800G DSP were shipped, while demand for 1.6T DSP might be over 20mn units; with actual shipments estimated at 16-17mn units. 800G DSP uses 7nm and 5nm nodes, while 1.6T requires 3nm technology that is facing insufficient capacity. The expert expects 1.6T DSP capacity to see a significant increase in 2H27E.
- For 2027E, the expert estimates 800G DSP shipment to reach 120-150mn units, not only coming from AIDC demand but also from the telecom market that is entering a replacement cycle. The expert estimates demand for 1.6T DSPs in 2027E at 60-70mn units, but the supply at 40-50mn units, with capacity and yield rates continuing to rise.

High-end DSP technology barriers and competition landscape

- 2026 marks the first year of large scale shipments of 3nm DSP (for 1.6T transceivers), and there are a few bottlenecks, according to the expert: 1) low yields; 2) some algorithm bugs that need to be fixed, i.e., FEC (forward error correction) algorithms; and 3) compatibility issues between high-speed DSP interfaces and network switches.
- The technical barriers for DSPs are high for newcomers, due to: 1) the manufacturing process requires advanced node technology (sub-7nm), and the cost for taping-out is very high; 2) technologies including signal processing algorithms, such as FEC, bit error rate, equalization technology, color temperature and dispersion compensation, are relatively challenging; and 3) power consumption management and thermal control are also difficult.
- Some second-tier players can also produce 800G DSPs, but they do not have a strong customer ecosystem. DSPs require both optical transceiver makers and end customers to participate in testing, and each testing cycle could last about two years. Therefore, end customers have high switching costs, and they tend to keep working with their existing suppliers, according to the expert.
- Some Chinese DSP manufacturers, such as Citrus Technology (unlisted) and Joywell Semiconductor (unlisted), are already working on 800G DSPs.

Coherent Lite DSP adoption

- Google (GOOG US, Not rated) uses 2.4T coherent-lite transceiver for its TPU and OCS network. The coherent-lite DSP is purposely customized from the traditional coherent DSP (used in telecom or long distance network) in order to be applied in the AIDC network. The coherent-lite DSP only uses a single fixed wavelength band (O-band), covering a 2-20km distance range.

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- The technical barriers of coherent-lite DSPs are lower than that of existing coherent DSPs, but higher than the standard PAM4 DSPs.

800G, 1.6T DSP price trend

- 800G DSP long-term contract (LTA) prices currently range USD45-70 and 1.6T DSP LTA price is at USD120-150 while retail prices are 10-20% higher, according to the expert.
- The expert thinks DSP price will slightly decline in 2027E with a higher capacity, with prices of 800G DSP dropping about 10% and a smaller extent for 1.6T DSP.

TIA / Driver development

- Leading TIA / Driver manufacturers have already achieved mass production of 200G per lane TIAs and drivers. 400G TIAs and drivers are still in development, and the main technology barriers are to achieve high linearity, low noise, and low power consumption simultaneously, according to the expert.
- As TIA and driver can use 28nm advanced node technology, Chinese manufacturers have achieved mass production of 100G TIA and drivers, while 200G is in the sample delivery stage. However, there are still gaps between domestic and overseas manufacturers including product stability and customer validation cycles.

Impact for electrical chips in NPO/CPO

- DSP would be removed for NPO (Near-field packaged optics) / CPO (Co-packaged optics) to reduce power consumption. After the DSP is removed, functions such as equalization, bit error and distortion correction will be partially handled by TIA/drivers and partially by switch chips. This places higher technology requirements on the linearity of TIA and driver.

Appendix A-1

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